

Long-Form Prose

Technical Writer

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1 The Challenge of PDF Conversion

Converting PDF documents to Markdown is fundamentally challenging because the two formats represent content in fundamentally different ways. PDF is a page description language that specifies the exact position, font, and size of each character on a page. It is designed for precise visual reproduction across different devices and platforms, prioritizing appearance over structure. Markdown, on the other hand, is a semantic markup language that describes the logical structure of content through simple syntax conventions like hash marks for headings, asterisks for emphasis, and pipe characters for tables.

The gap between these two paradigms means that any converter must bridge the divide between visual presentation and logical structure. This requires the converter to reverse-engineer the author's intent from the visual layout, inferring paragraph boundaries from spacing, heading levels from font sizes, and list items from indentation and marker characters. This process is inherently heuristic and imperfect, because the same visual appearance can result from different logical structures, and different tools may produce subtly different PDF output for the same source document.

2 Our Approach

Our six-layer pipeline approaches this challenge by separating concerns into distinct processing stages. The first layer extracts raw character data from the PDF, including precise position coordinates, font information, and text content. The second layer assembles these characters into words, lines, and blocks using adaptive gap analysis that adjusts thresholds based on the local font size. The third layer performs layout analysis to detect headers, footers, columns, and reading order. The fourth layer applies semantic classification to identify headings, paragraphs, lists, and other structural elements. The fifth layer handles table extraction using specialized algorithms. The sixth and final layer generates the Markdown output.

3 Content Accuracy as the Primary Goal

Throughout the entire pipeline, our primary goal is content accuracy rather than formatting fidelity. We prioritize ensuring that every word, number, and symbol from the source PDF

appears in the output Markdown, even if the formatting is not perfectly replicated. This design decision reflects the intended use case: converting documents for consumption by AI and language models, where content completeness matters more than visual appearance.